

Introduction

To support our forecasting analysis, the following tables and figures summarize the main parts of our modeling process, including model selection, feature engineering, and performance evaluation. Table 1 shows the different model types we explored and the hyperparameters used in the final versions, while Table 2 explains the engineered features included in the dataset and why we chose to include them in our final dataset. Table 3 compares model performance across approaches using Root Mean Squared Error (RMSE) as the evaluation metric, and Figure 1 provides a screenshot of our best Kaggle submission as an external benchmark of overall model performance.

Results

Table 1: Explored Models & Hyperparameters

Model Name	Hyperparameters Used
LightGBM	objective = Tweedie, boosting_type = gbdt, learning_rate = 0.0419, num_leaves = 91, min_data_in_leaf = 85, tweedie_variance_power = 1.2766, lambda_l1 = 0.8683, lambda_l2 = 0.3598, feature_fraction = 0.6988, bagging_fraction = 0.8689, bagging_freq = 1, best_iteration = 152.
Random Forest	n_estimators = 300, max_depth = 20, min_samples_leaf = 20, max_features = sqrt
Ridge Regression	alpha = 0.5, fit_intercept = true
GRU	sequence length = 28, GRU units = 32, dropout = 0.10, dense layer = 32, optimizer = Adam, learning_rate = 0.001, epochs = 15, batch_size = 51, early stopping patience = 3

Table 1: Explored Models & Hyperparameters: List of the models explored and the key hyperparameters evaluated in this competition. For LightGBM, the selected settings were objective = Tweedie, boosting_type = gbdt,

learning_rate = 0.0419, num_leaves = 91, min_data_in_leaf = 85, tweedie_variance_power = 1.2766, lambda_l1 = 0.8683, lambda_l2 = 0.3598, feature_fraction = 0.6988, bagging_fraction = 0.8689, bagging_freq = 1, and best_iteration = 152. For Random Forest, the final settings were n_estimators = 300, max_depth = 20, min_samples_leaf = 20, and max_features = sqrt. For Ridge Regression, the selected settings were alpha = 0.5 and fit_intercept = true. For GRU, the selected settings were sequence length = 28, GRU units = 32, dropout = 0.10, dense layer = 32, optimizer = Adam, learning_rate = 0.001, epochs = 15, batch_size = 51, early stopping patience = 3.

Table 2: Explored Features & Rationale of Feature Engineering

Feature Name	Category	Description / Rationale	RMSE
Baseline Features			
item_id	Categorical	Unique product identifier; allows the model to learn item-level demand patterns.	Overall: 2.1619
dept_id, cat_id	Categorical	Department and category groupings; help capture shared demand trends across product lines.	
store_id, state_id	Categorical	Store and state location identifiers; capture geographic demand differences across regions.	
dow, month	Calendar	Day of week and month capture weekly and seasonal demand cycles in retail sales.	
sell_price, price_change, is_promo	Price / Promo	Selling price, daily price change, and promotion flag signal consumer price sensitivity and promotional lift effects.	
lag_7, lag_28	Lag & Rolling	Sales from 7 and 28 days ago; capture short-term weekly and monthly demand momentum.	
rmean_7, rmean_28	Lag & Rolling	Rolling 7-day and 28-day means of lagged demand; smooth out noise and capture underlying demand trends.	
Added Features			
snap_flag	Calendar	Flags SNAP benefit days by state (CA, TX, WI); SNAP days systematically boost food sales and are known in advance.	Combined with baseline feature: 2.1322
is_event	Calendar	Binary flag for any holiday or promotional event; captures demand spikes on special occasions and is known in advance.	
lag_56	Lag & Rolling	Sales from 56 days ago; captures bi-monthly demand cycles and is the minimum safe lag for a 28-day forecast horizon.	

Table 2: Explored Features & Rationale of Feature Engineering. Summary of all engineered features used in the final LightGBM model, grouped into baseline and added features. Baseline RMSE (combined): 2.1619. RMSE values for added features reflect each feature's standalone performance; the best combined result with snap_flag, is_event, and lag_56 achieved an RMSE of 2.1322, a 1.4% improvement over the baseline.

Table 3: Model Performance Comparison

Model Name	Performance Metric (RMSE)
LightGBM	2.0025
Random Forest	2.0802
Ridge Regression	2.1180
GRU	2.2583

Table 3: Model Performance Comparison. Comparison of model performance using RMSE, where lower values indicate better forecasting accuracy. LightGBM performed best with an RMSE of 2.0025, outperforming Random Forest (2.0802), Ridge Regression (2.1180), and GRU (2.2583).

Figure 1: Screenshot of Best Model Performance (Kaggle):

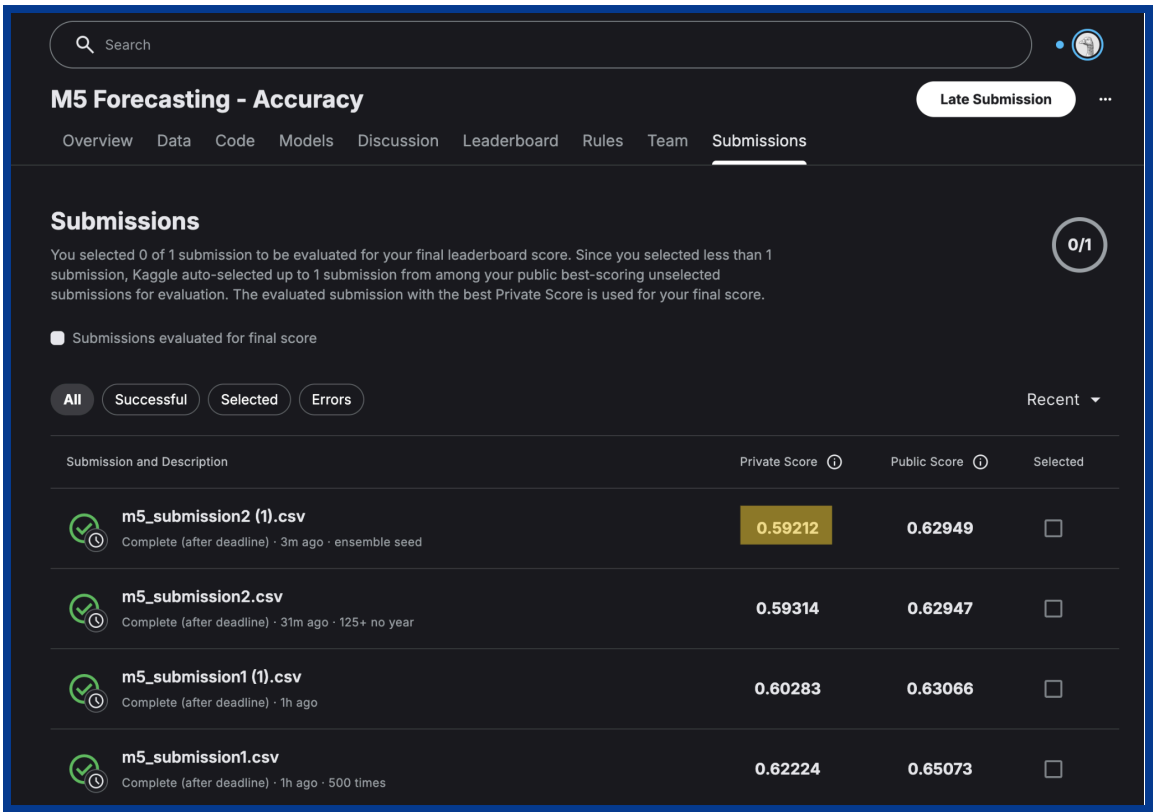


Figure 1: Screenshot of Best Model Performance. Kaggle M5 Forecasting competition submission page showing the model’s results, with the best submission achieving a private score of **0.59212** and a public score of **0.62949**.